Harsh Jha

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PROFILE

- AI-focused Software Development Engineer with **2.5 years of experience** building production-grade LLM applications, agentic AI workflows, scalable backend systems, and cloud-native microservices.
- Hands-on experience designing and integrating **LLM and agentic AI workflows** using LangGraph, LangChain, RAG pipelines, tool calling, vector search, AWS Bedrock, prompt engineering, and structured outputs.
- Strong backend and systems foundation across **Python, Go, TypeScript, REST APIs, PostgreSQL, Redis, Docker, Kubernetes, AWS, and Azure**, with experience taking features from design through production.
- Open-source maintainer of **VitaTrack.ai**, an AI-powered healthcare platform for medical document extraction, report summarization, health-record organization, and intelligent health-data workflows.

EDUCATION

Indian Institute of Technology Indore (IIT Indore) **Indore, India**
Bachelor of Technology (B.Tech.) in Civil Engineering *2020 – 2024*

TECHNICAL SKILLS

- **AI / GenAI:** AI Agents, LangChain, LangGraph, RAG, Tool Calling, LLMs, Agentic AI, Prompt Engineering, Function Calling, LLM Integration, AI Workflow Automation
- **AI Engineering:** Vector Search, Agent Orchestration, Retrieval Pipelines, Embeddings, Context Engineering, Structured Outputs, Model Integration, AI Evaluation
- **Programming:** Python, Go, TypeScript, JavaScript, C++
- **Backend:** Microservices, FastAPI, Gin, Fiber, Node.js, Express.js, REST APIs, gRPC, API Gateway
- **Data:** PostgreSQL, MongoDB, MySQL, SQL Server, Redis
- **Cloud & AI Services:** AWS, AWS Bedrock, Azure, AWS Textract, Oracle Cloud Infrastructure
- **DevOps & Infrastructure:** Docker, Kubernetes, GitHub Actions, CI/CD, Containerized Deployment
- **Architecture:** System Design, Distributed Systems, Event-Driven Architecture, Microservices, Clean Architecture, SOLID Principles
- **Security:** JWT, RBAC, API Validation, Secure API Design

WORK EXPERIENCE

Software Development Engineer - 2 **May 2024 – Present**
SimplyPhi *Bangalore, India*

- Designed and built scalable backend services using **Go, Python, TypeScript, REST APIs, PostgreSQL, and microservices** for production business workflows.
- Developed production-ready APIs for **authentication, reporting, financial workflows, property operations, dashboard analytics, and business automation**, supporting multiple application workflows.
- Engineered AI-powered workflows using **LangChain, LangGraph, AI agents, LLM APIs, RAG pipelines, prompt engineering, tool calling, and structured outputs**.
- Integrated **LLM and AI services** into business applications for document understanding, report generation, data extraction, summarization, and intelligent workflow automation.
- Built and deployed cloud-native services using **Docker, Kubernetes, AWS, Azure, GitHub Actions, and CI/CD**, supporting reliable application delivery across environments.
- Implemented secure application workflows using **JWT authentication, RBAC, API validation, structured logging, and centralized error handling**.
- Improved backend scalability and maintainability through **Clean Architecture, modular service design, optimized database queries, caching, and reusable service components**.
- Owned features end-to-end across **requirements analysis, system design, development, testing, deployment, monitoring, debugging, and production support**.

PROJECTS

VitaTrack.ai – AI Medical Report Platform | Go, Python, PostgreSQL, AWS Bedrock, Microservices GitHub

- Developing an open-source AI healthcare platform for managing **medical reports, prescriptions, symptoms, medications, health records, and daily health notes**.
- Built secure **Go backend APIs** and structured healthcare data workflows using **PostgreSQL** for medical record management.
- Integrated **AWS Textract** to extract text and structured data from medical reports, prescriptions, and healthcare documents.
- Integrated **AWS Bedrock and LLMs** with prompt engineering to summarize medical reports and transform unstructured health data into structured, actionable information.
- Designed AI workflows for **medical document analysis, health-history tracking, report processing, and patient record organization**.
- Built supporting **Angular** interfaces for document uploads, health logs, dashboards, and medical data management.

AI Agent Workflow System | Python, LangGraph, LangChain, RAG, Tool Calling GitHub

- Engineered multi-step **AI agent workflows** using Python, LangChain, LangGraph, and LLM APIs for intelligent automation and structured decision-making.
- Designed stateful agent workflows for **document processing, summarization, API integration, tool execution, and structured response generation**.
- Implemented **tool calling, prompt templates, reusable chains, memory, state management, and conditional workflow execution** for agent-driven automation.
- Integrated AI agents with **REST APIs and external tools** to execute tasks, process information, and support real-world business workflows.

CERTIFICATIONS

- **Introduction to LangChain - Python** — LangChain Academy
- **Introduction to Agent Skills** — Anthropic
- **Oracle Cloud Infrastructure 2025 Certified** — Oracle

ACHIEVEMENTS

- Achieved a **LeetCode Contest Rating of 1663**, placing in the **top 6% globally**.
- Open-source maintainer of **VitaTrack.ai**, an AI-powered healthcare platform focused on medical document processing and intelligent health-data workflows.